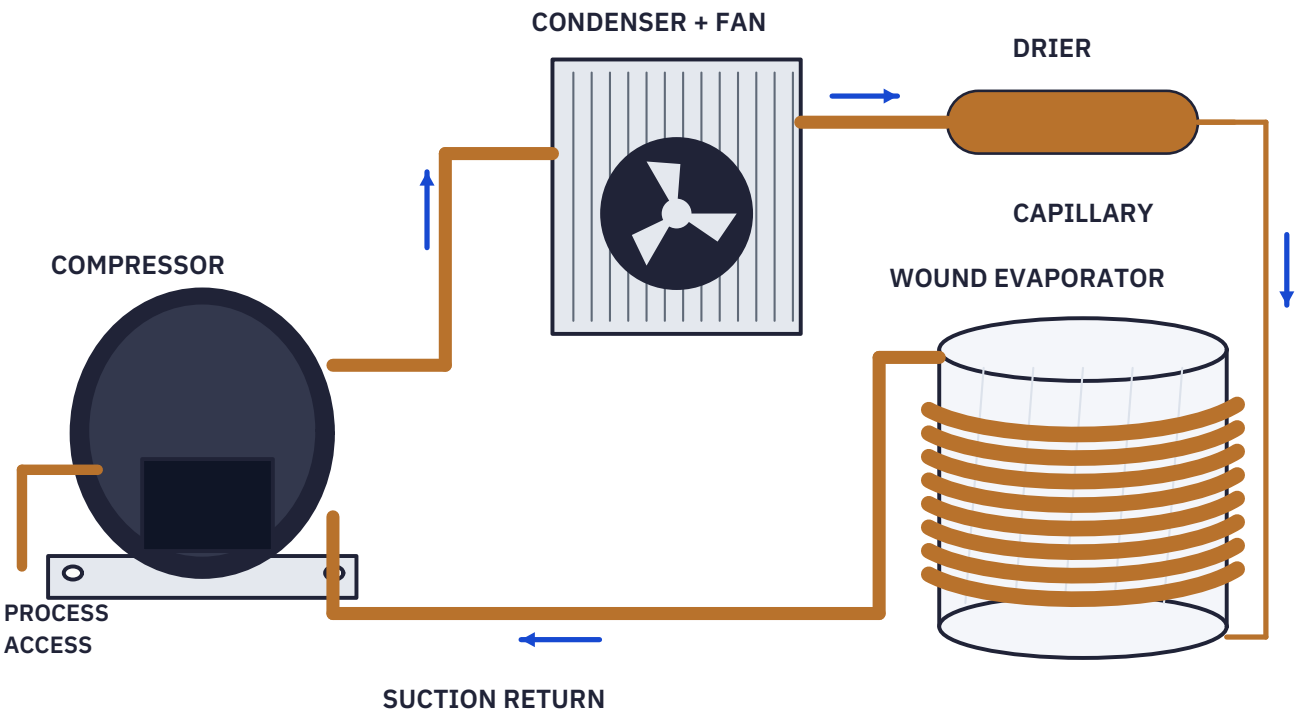
Build the cold loop

Wind the evaporator, transfer the donor circuit, and prove the installed refrigeration assembly.



1 Follow the physical sequence

Coil work: pages 4-8. Donor prep and circuit work: 2-3 and 9-19. Mounting and commissioning: 20-27.

2 Read the pictures

Coral marks the current work. Copper is refrigerant tube. Blue marks a tool, gauge or movement. Pictures are schematic; printed dimensions govern.

3 One valve for life

The BPV31 goes onto the process tube once. Venting, the argon purge, vacuum and recharge all pass through its flare port, and it stays as the service point.

CHECK

Argon flows from the first cut until vacuum starts; the factory drier stays in service throughout.

Read the donor's label

One label decides the session: the donor must run R600a, and its factory charge is the number the recharge target builds from.

APPLIANCE RATING LABEL

MODEL
Read exact model

REFRIGERANT
R600a

CHARGE
Copy actual grams

VOLTAGE / Hz
Copy actual supply

→

REFERENCE BASELINES

Unit A / HZB-12/Q	15 g
Unit B / EFIC117-SS	23 g

The donor's label governs.

Cast-stamps 48.5-2 / 45
are not charge masses.

1 Photograph the real labels

Record appliance model, refrigerant, factory charge, compressor voltage, frequency and current ratings. Keep each donor's data with its parts.

2 Verify R600a

The type must be R600a. Any HFC donor (R-134a, R-410a) is a full stop: this procedure does not apply. Record the factory charge: 15 g Unit A, 23 g Unit B.

3 Mark the assembly

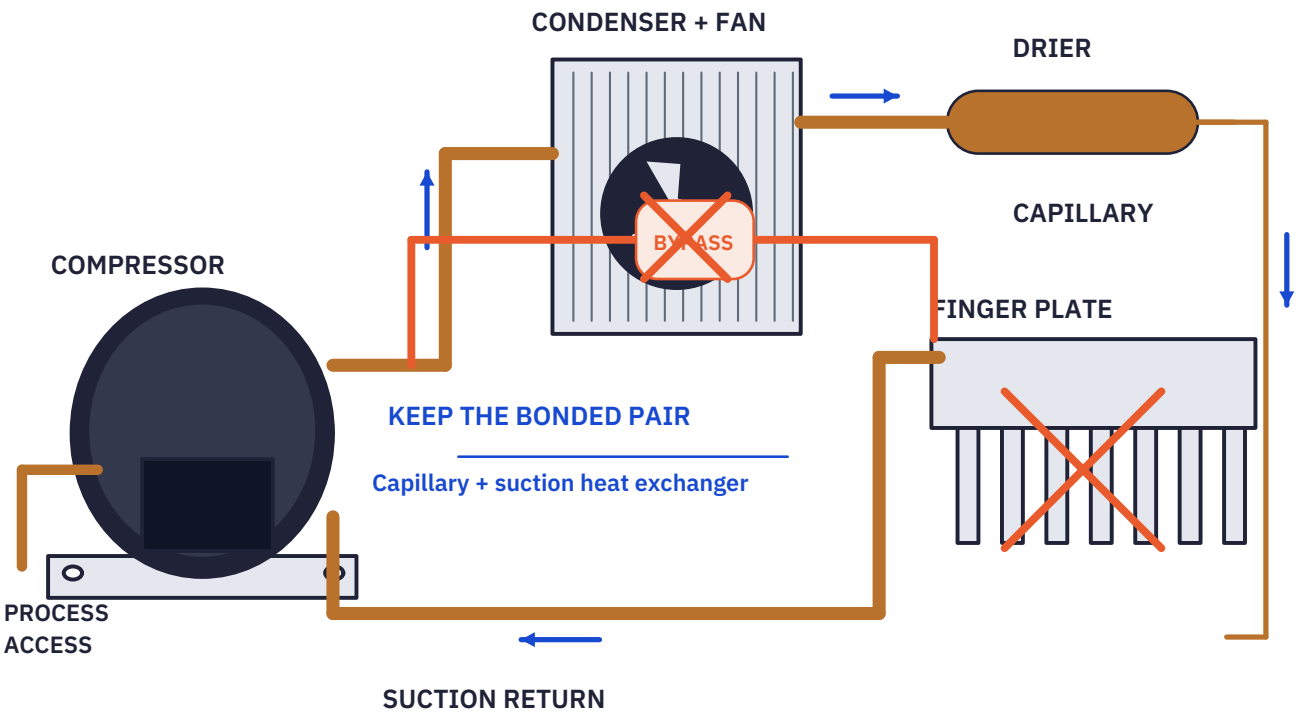
Tag Unit A or Unit B and its compressor. Do not infer charge from the cast-stamps or borrow Unit A's electrical lead colors for Unit B.

DONE WHEN

The label reads R600a and the factory charge is recorded: 15 g (A) or 23 g (B).

Trace what stays

Walk every tube while the donor is still intact. Mark the two cuts before dismantling.



1 Keep the useful assembly

Compressor, condenser/fan, original terminal cover, grommets and the bonded capillary/suction pair stay together.

2 Remove the ice-making path

The finger-plate evaporator and the hot-gas bypass solenoid, line and tee come out. Remove the valve entirely, or leave it in place and never energize it.

3 Keep the factory drier

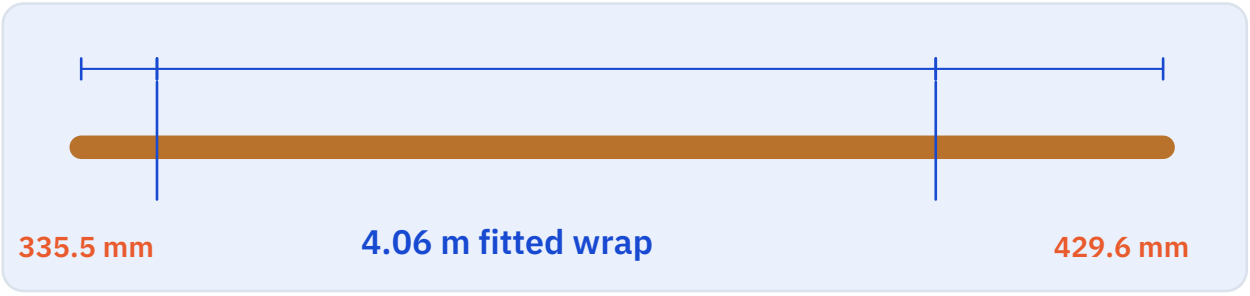
The drier, its brazed-on capillary, the cap-tube helix and the bonded suction pair stay together as one preserved subassembly, at factory length.

CHECK

Both donors' topologies are verified. Nothing upstream of the evaporator-inlet cut is disturbed.

Reserve both coil tails

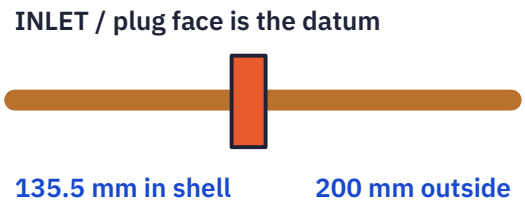
Cut the dry coil stock with a tubing cutter, with enough copper for the wrap and each separate routed tail.

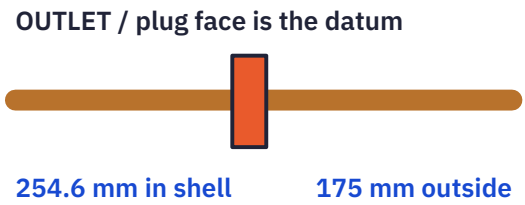


Inlet allowance

Outlet allowance

Total cut / about 4.85 m (15.92 ft)





length picture is schematic

1 Use the specified stock

GOORY ACR copper, 1/4 in OD x 0.031 in wall. Keep its bore clean and capped. Do not saw refrigerant tubing.

2 Mark the wrap and tails

Current nominal stock is about 4.85 m total: 4.06 m fitted wrap plus 335.5 mm inlet and 429.6 mm outlet allowances.

3 Keep the allowances intact

Each allowance includes the in-shell route and protrusion. The longer outlet route still leaves the shorter outside working length.

CHECK

The two tails are labeled and the ends are round, square, burr-free and protected from dirt.

Wind the copper coil

Use the printed mandrel groove to place a single layer of tubing without flattening the bore.



1 Start on the correct tail

Reserve the inlet allowance, register the first wrap in the groove, and keep the tail on its assigned side.

2 Follow the helix

Wind 9.687 wraps at 12.33 mm pitch over 119.4 mm. The mandrel is 123 mm OD. Keep steady contact without crushing or kinking the tube.

3 Remove without bending a tail

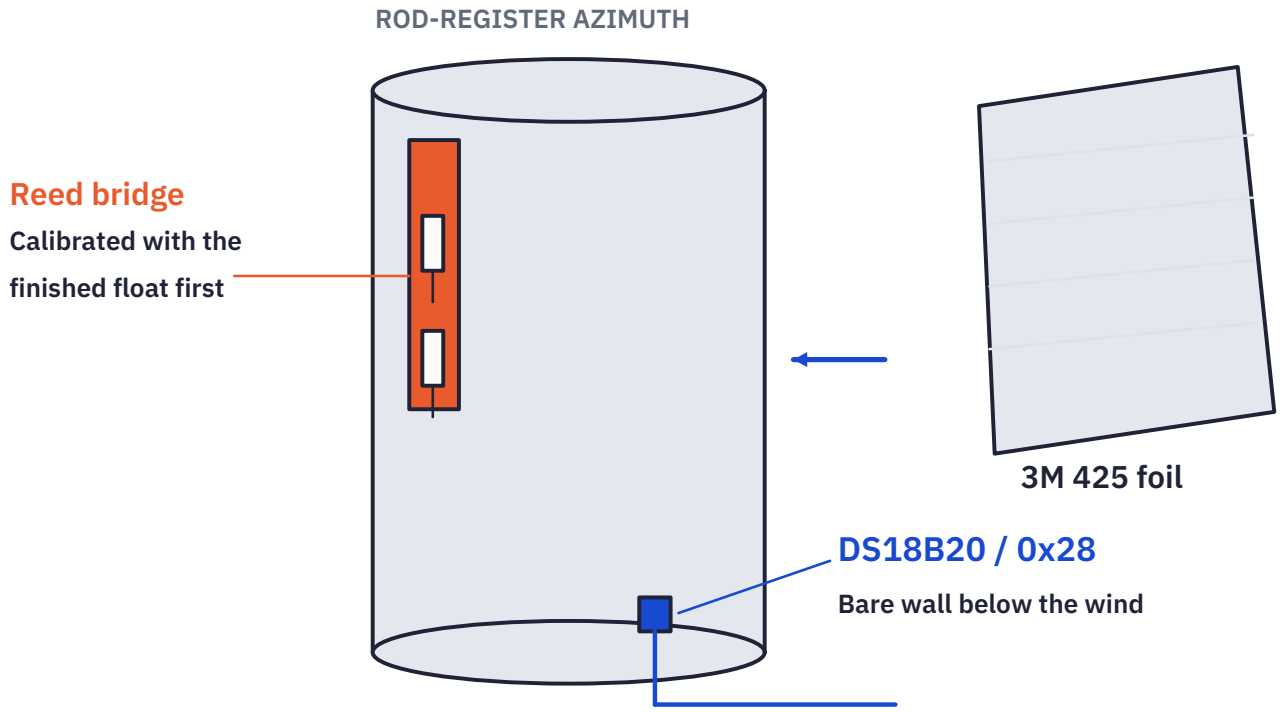
Slide the coil axially off the mandrel. Inspect every turn and both exits; a flattened or sharply creased tube is not usable.

DONE WHEN

The free coil holds one layer and the inlet/outlet tails leave on opposite assigned lanes.

Dress the bare wall

Install and prove everything that needs direct steel contact before foil, coil or foam covers it.



1 Bond the wall probe

Tape the DS18B20, family 0x28, flat against bare carbonator steel in the band below the wind. Insulate its leads and route them upward.

2 Set the reed bridge

Use the rod-register azimuth. Calibrate the finished float and both directional liquid crossings before fixing the final bridge position.

3 Apply the foil skin

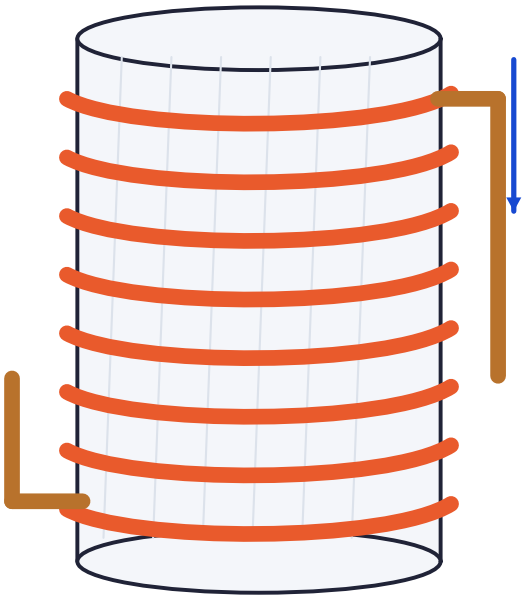
Apply continuous 3M 425 foil between steel and coil. Preserve the proven sensor positions, lead insulation and the bridge's crossing envelope.

DONE WHEN

Probe and reed checks are recorded while every contact point is still reachable.

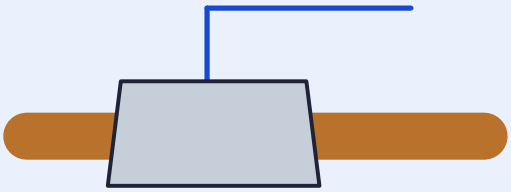
Transfer and bond the coil

Slip the wound coil over the foil-dressed 127 mm carbonator, then place the freeze-protect probe.



127 mm carbonator

SUCTION-END PROBE



DS18S20 / 0x10

Flat on copper at the high / outlet end

Secure before foam hides it.

1 Slide the coil into place

The wound coil is undersize to hold against the tank. Ease it on without kinking a tail, dislodging a reed or abrading insulated leads.

2 Add the coil probe

Tape the DS18S20, family 0x10, flat against suction-end copper at the high/outlet end. The wall probe stays on steel below the wind.

3 Secure and prove

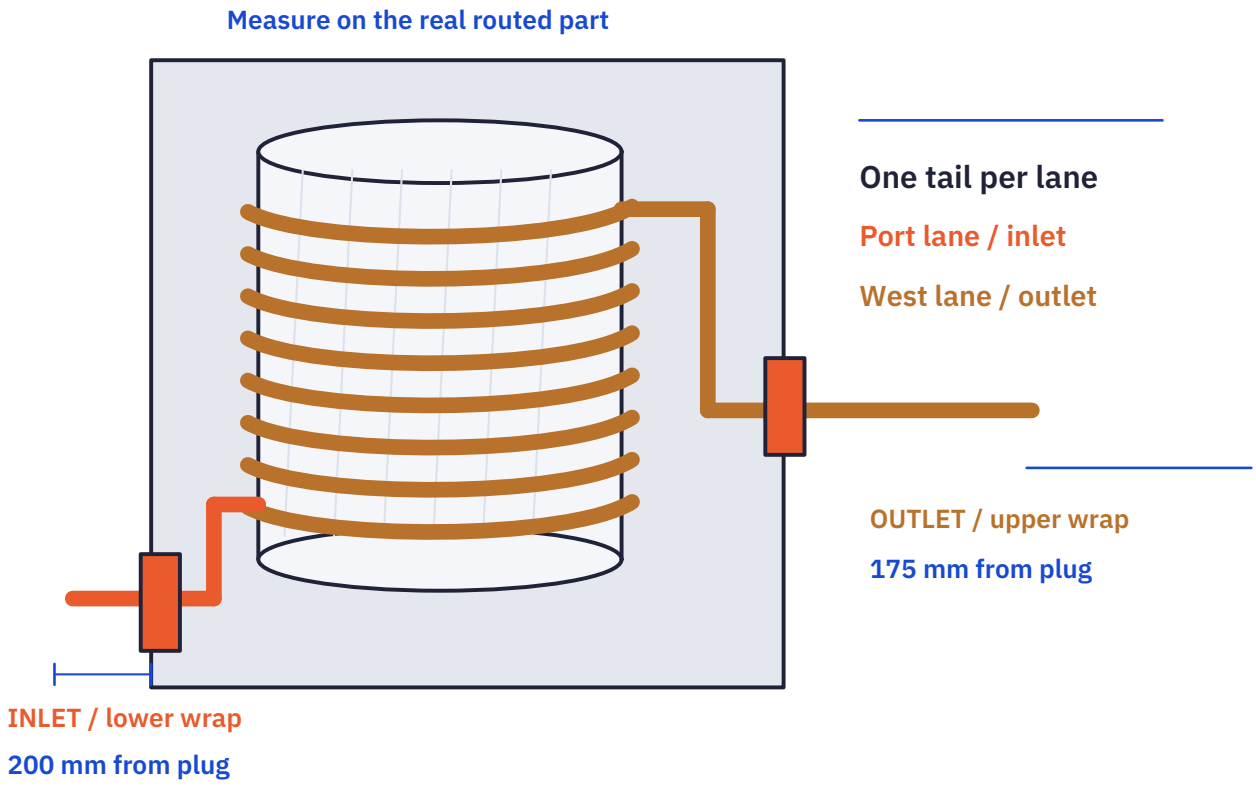
Bond the thermal contact with foil. Prove both probe families and lead continuity before the mold guide's pre-pour check hides the surfaces.

CHECK

Two different probe families: 0x28 on steel; 0x10 on copper. Neither probe floats in foam.

Route the two exits

Lower the dressed carbonator into the open shell with one refrigerant tail in each assigned lane.



1 Clock by the tails

The lower/inlet tail takes the port lane. The upper/outlet tail takes the west lane. Avoid torsion where either tail leaves the wrap.

2 Measure from each plug face

Leave 200 mm inlet and 175 mm outlet outside the plug. Measure the physical routed assembly; do not cut away the in-shell part of an allowance.

3 Handoff to the mold guide

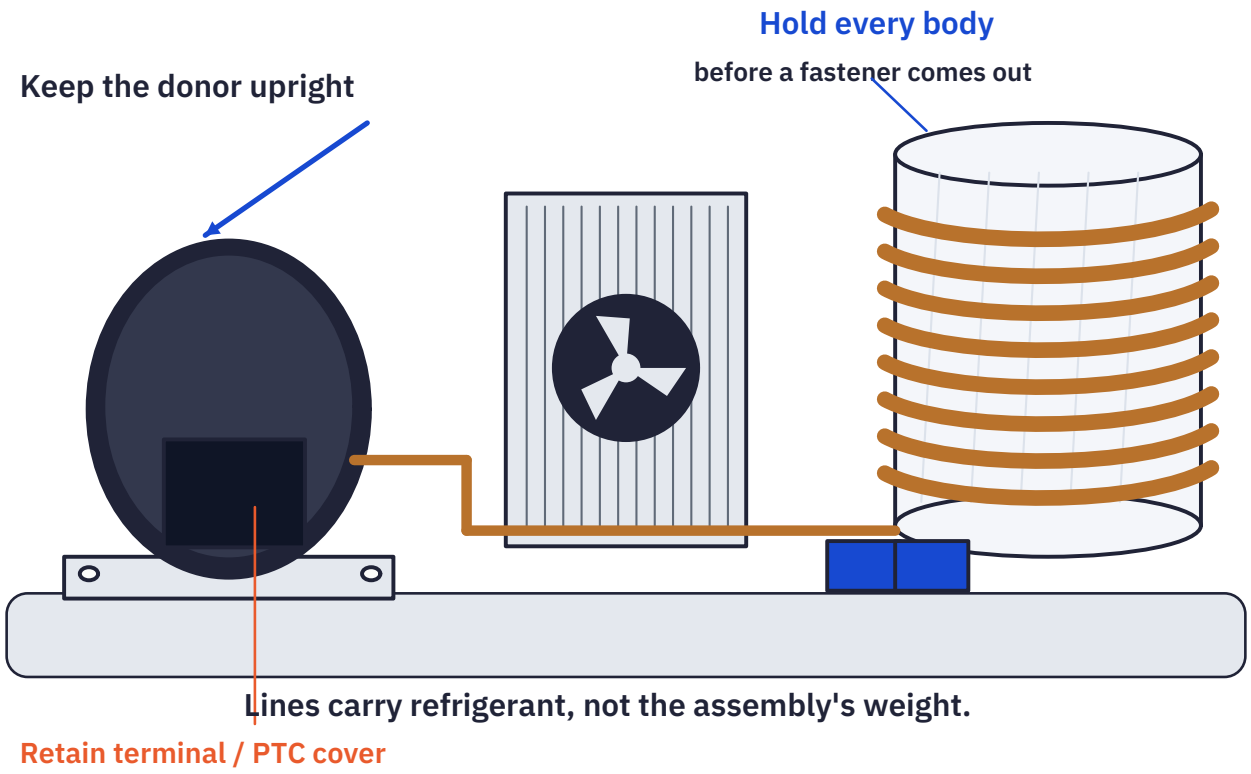
The copper plugs close their elongated wall slots, with sensor leads protected. Follow the mold guide for pre-pour inspection and cured closure.

DONE WHEN

Both protrusions are recorded and the cured core can be supported with no load on the tube exits.

Support before teardown

Remove only the donor casing and supports that prevent access, while keeping the sealed circuit undamaged.



1 Unplug and identify the leads

Keep mains isolated. Photograph the factory-external compressor and fan connections. Retain the terminal/PTC cover and its original retention.

2 Support each body

Keep the compressor upright. Support the condenser, fan and evaporator before undoing their fasteners. Copper tubing must not hold their weight.

3 Preserve the useful details

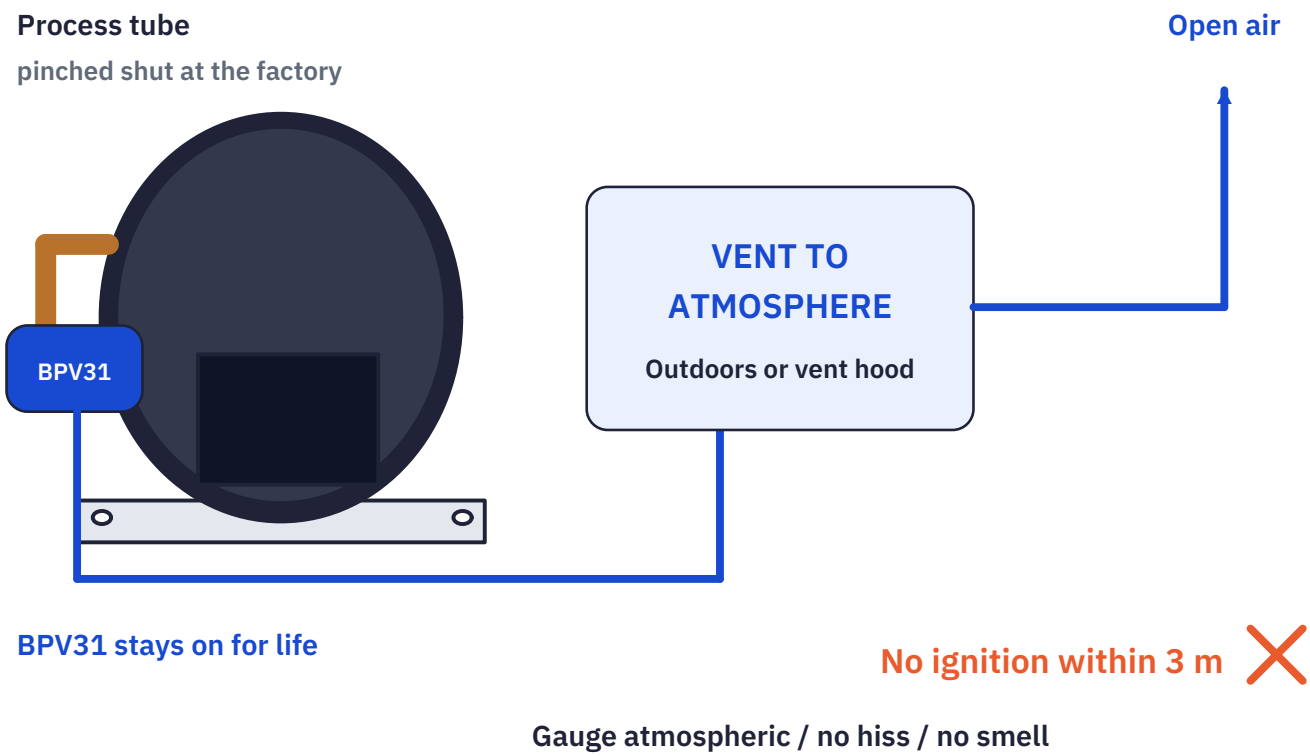
Retain grommets and mounting hardware for inspection. Protect condenser fins, the bonded capillary pair and the process stub. Keep Unit B lead identity open until recorded.

CHECK

A missing, cracked, altered or loose terminal cover fails the assembly before power is applied.

Vent the factory charge

The loop opens for the first time through a valve that never comes off. Decompress fully before any flame.



1 Find the process tube

The short copper stub off the compressor, pinched and brazed shut at the factory. It connects to nothing else.

2 Clamp the BPV31 on for life

Clamp the saddle on and pierce. It stays as the single permanent service point: venting, argon, vacuum and recharge all use its flare port.

3 Vent outdoors and confirm

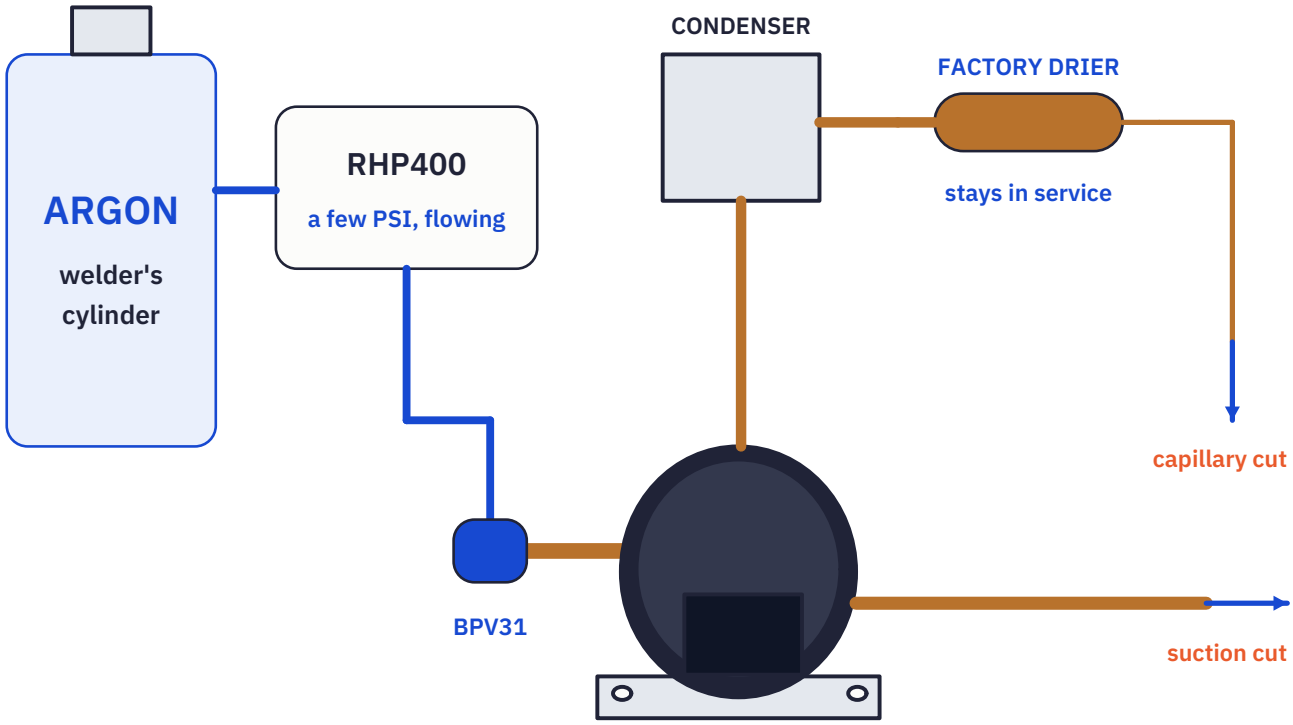
Open the valve outdoors or under a vent hood, with no ignition source within 3 m. This vents the household donor; it does not authorize venting a rebuilt unit.

DONE WHEN

Gauge reads atmospheric, no further hiss, no smell at the valve. Only then is anything cut or heated.

Start the argon purge

Argon first, cuts second. One continuous low-pressure flow guards the whole loop-open period.



Argon enters at the BPV31 and leaves by both open cuts.

1 Rig it to the BPV31 flare

Uniweld RHP400 on the welder's argon cylinder, then the flared 1/4 in ACR stub, Joywayus flare nut and charging hose onto the BPV31 flare port.

2 Flow a few PSI before cutting

Open the BPV31 and start low-pressure argon, flowing, not static. Vented is not empty: R600a lingers in the compressor oil and in low spots.

3 Never break the flow

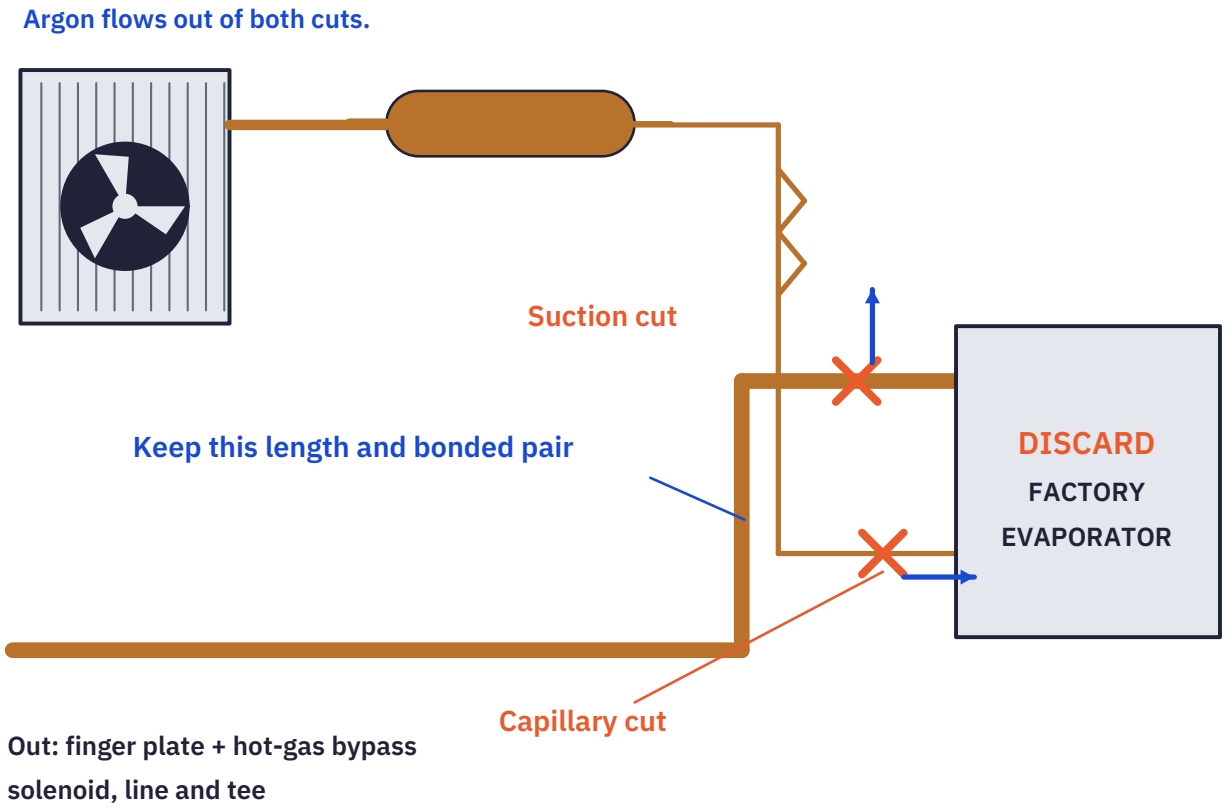
Argon runs from the first cut until vacuum begins on page 16. It is the fuel sweep and the factory drier's dry blanket at the same time.

CHECK

A lapse in flow lets the factory drier's desiccant take on moisture and invites capillary icing.

Cut at the evaporator

Remove the ice-making evaporator with argon flowing, leaving the drier, capillary and bonded heat exchanger at factory length.



1 Cut the suction line

Between the evaporator outlet and the compressor inlet, close to the evaporator, with the RIDGID tubing cutter. Argon leaves here.

2 Cut the cap tube at the evaporator

At its evaporator-inlet end, with the Mastercool capillary cutter. Drier, cap tube and helix stay untouched at factory length; argon leaves here too.

3 Strip the ice-making path

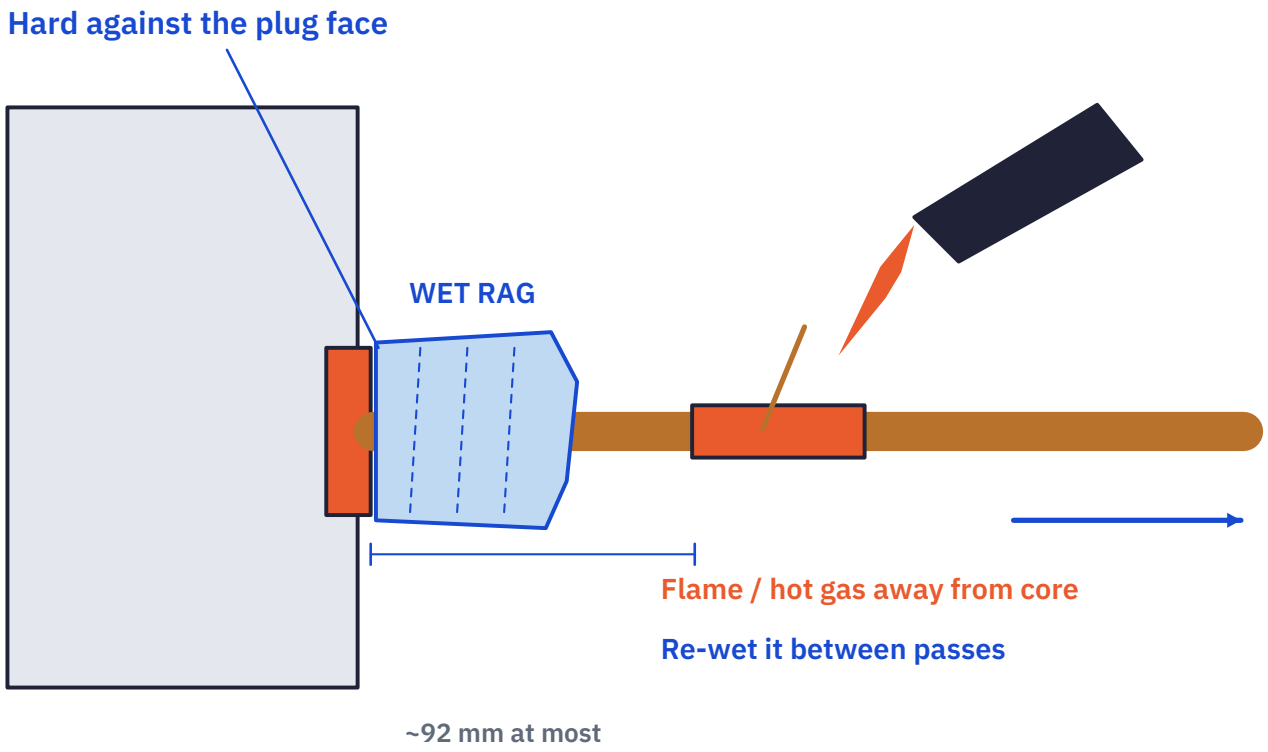
The finger-plate evaporator and the hot-gas bypass solenoid, line and tee come out. The bonded capillary/suction pair stays intact on the compressor side.

DONE WHEN

Both cuts are open with argon flowing out of each; drier, cap tube, helix and bonded pair are at factory length.

Wet rag on each stub

Both joints sit on copper that runs back into a printed PETG plug with cured foam behind it. Protect it before each braze.



1 Soak and wring a rag

Cotton shop rag, heavy with water but not running. Do this immediately before each braze; a rag that has been through one braze is dry.

2 Wrap it hard against the plug

On the core side of the joint, covering as much of the stub as it reaches. The rag is the heat sink: wrap toward the plug, not the joint.

3 Keep it wet; aim away

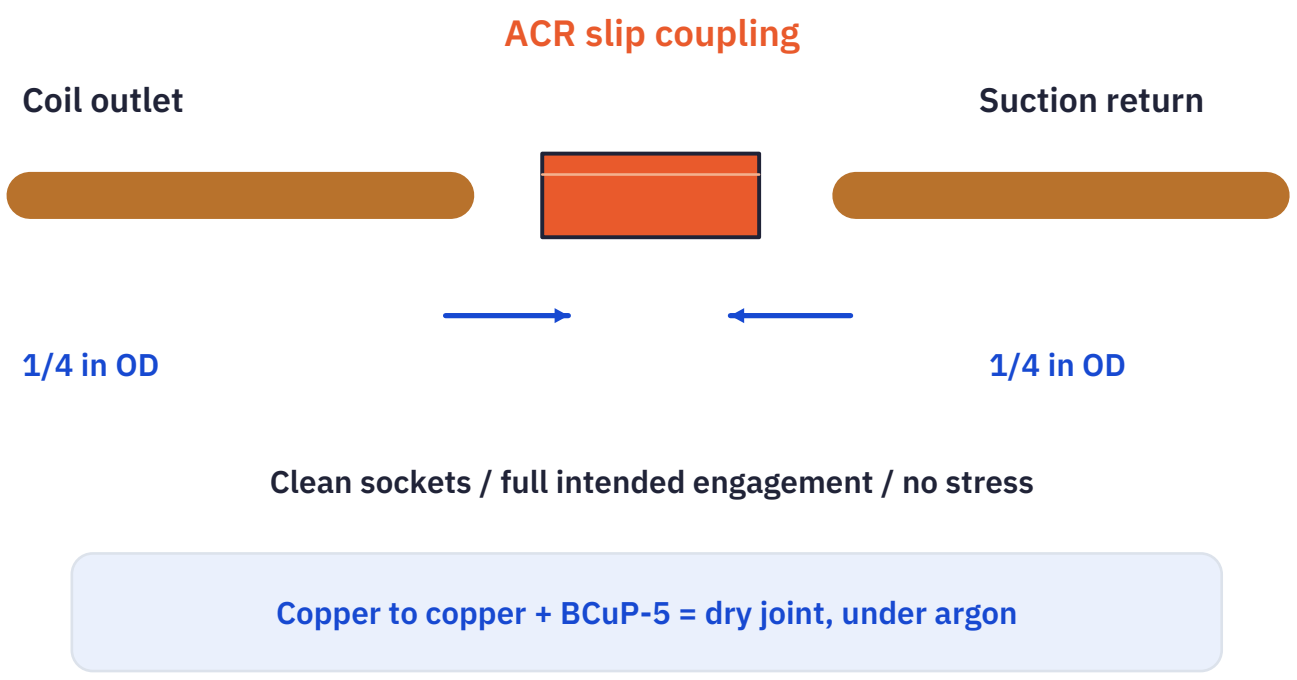
Re-wet it between passes. Aim flame and hot gas away from the core, toward the compressor side, never along the stub toward the plug.

CHECK

After each braze: gloss, sag, dimpling or a scorched-foam smell at the plug face means it is compromised.

Braze the suction coupling

Join the upper coil outlet to the factory suction line with an ACR-grade 1/4 in slip coupling, under the flowing argon.



1 Bright copper first

Scotch-Brite both tube ODs and the coupling sockets until bright. Keep grit and swarf out of the bore.

2 Stage the joint

The coil-outlet stub leaves the wall on the west lane at the factory suction cut. Slide the slip coupling over both 1/4 in ends; no adapter.

3 Braze dry under argon

Wet rag on the stub first (page 13). BCuP-5 self-fluxes on copper, so no flux goes into the sealed loop. Flame away from the core.

DONE WHEN

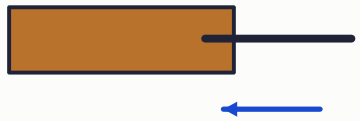
A complete fillet ring at both coupling mouths, argon never interrupted, plug face unmarked.

Pinch-swage the capillary

No reducer: the 1/4 in coil-inlet stub collapses onto the cap tube and is brazed under the flowing argon.

1

SLIDE THE CAP TUBE IN

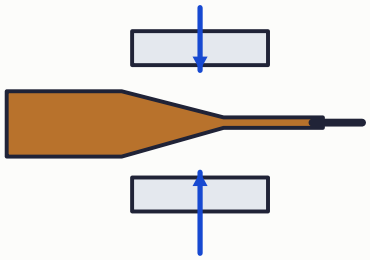


No reducer fitting

Factory length

2

FLATS, NOT BITES

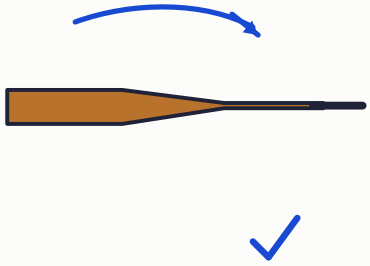


Knipex 86 01 180

Smooth parallel jaws

3

TURN, SQUEEZE, BRAZE



60 degrees per turn

Braze dry, no flux

1/4 in coil stub onto the ~0.031 in cap tube, under flowing argon.

1 Slide the cap tube in

The cap tube cut on page 12 goes into the coil-inlet stub, which leaves the front wall on the port lane.

2 Swage with the Knipex

Knipex 86 01 180 smooth parallel jaws, flats not bites: squeeze, rotate 60 degrees, squeeze again, until the stub has closed evenly onto the cap tube.

3 Braze dry with a fresh rag

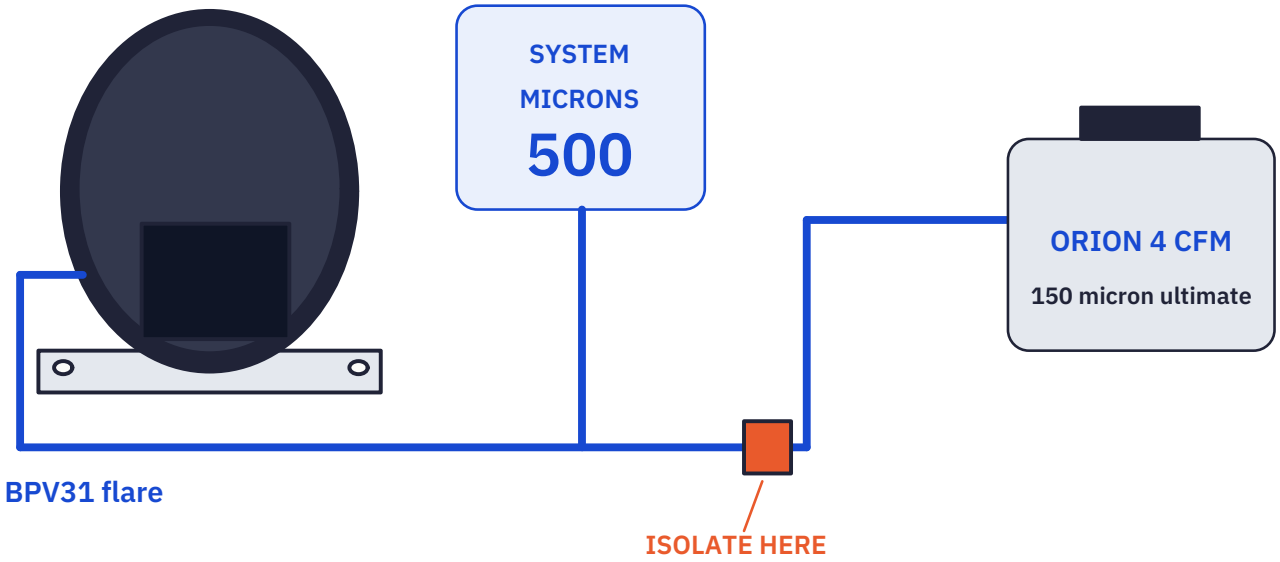
A fresh wet rag hard against the plug face; page 14's rag is dry. Braze the swaged mouth with no flux, then check the plug face.

CHECK

The cap tube stays at factory length. If it ever changes substantially, a refrigeration tech recalculates it.

Pull vacuum

First proof of the brazes, and the handoff: argon stops only because the pump takes over.



15 min on the pump, then valve off:

no rise for 15 min = tight.

1 Argon off, manifold on

With every braze done, stop argon at the RHP400 and close the BPV31. Swap the argon hose for the manifold's 1/4 in SAE flare; reopen the BPV31.

2 Pull to 500 microns

Orion 4 CFM pump (150 micron ultimate) through the Orion manifold. Reach 500 microns or below and hold at least 15 minutes on the pump.

3 Valve off and watch

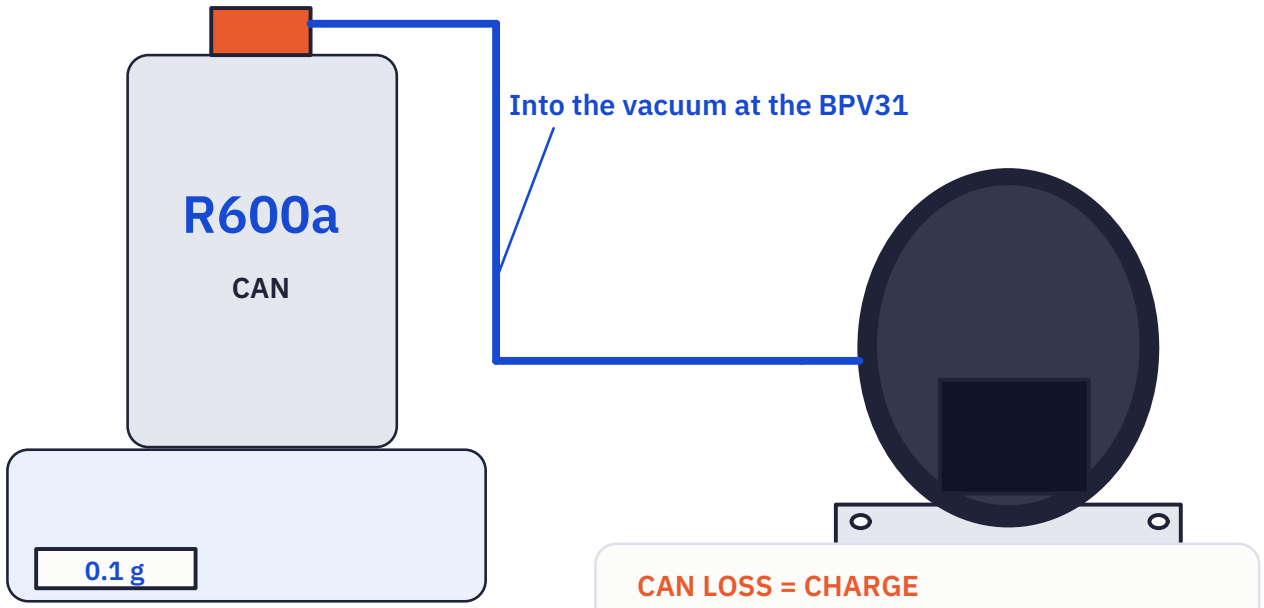
Isolate the pump. No rise for 15 minutes means tight. A rise is moisture (pump longer) or a leak (find it and fix it).

DONE WHEN

500 microns or below, 15 minutes pumping, then 15 minutes valved off with no rise.

Meter the recharge

The scale is the meter: what the can loses, the loop gains.



Can + valve on scale

Start at factory mass + 5-15 g; iterate 1-2 g.

1 Can on the scale, tared

Enviro-Safe R600a can on the Smart Weigh Pro (0.1 g). Plumb can to manifold to the BPV31 flare port; open the BPV31.

2 Charge into the vacuum

Open the can valve. R600a flows in on its own vapor pressure and the readout falls. Close at the target drop: the can's lost grams are the charge.

3 Start above factory

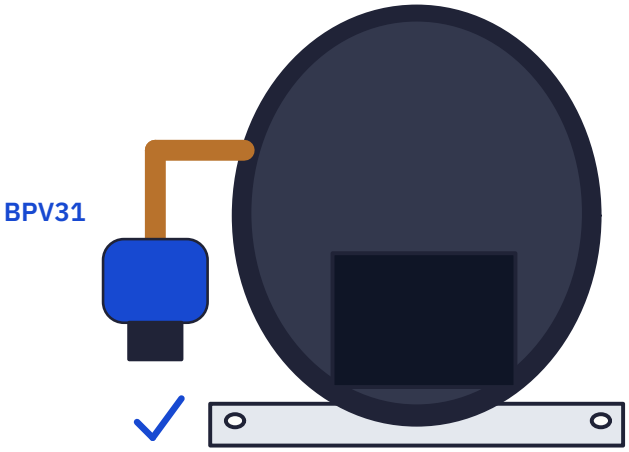
The wound coil holds more than the finger plate. Start at factory mass (15 g A / 23 g B) plus 5-15 g; iterate 1-2 g against frost pattern and suction superheat.

CHECK

Charge within +/-1 g of the working target. The target is settled on first run-up (procedure open item 1).

Cap the BPV31 for life

The valve that vented the donor stays on as the appliance's only service point.



Closed and capped

Clamped on for life

ONE SERVICE POINT

- Vent
- Argon purge
- Vacuum
- Recharge
- Later service

Record net grams on the unit

1 Button up

Close the manifold valve and the BPV31, disconnect the manifold and cap the BPV31 flare port.

2 Leave the valve on

The BPV31 stays clamped on the compressor process tube as the single permanent service-access point for vacuum, recharge and later service.

3 Record the charge

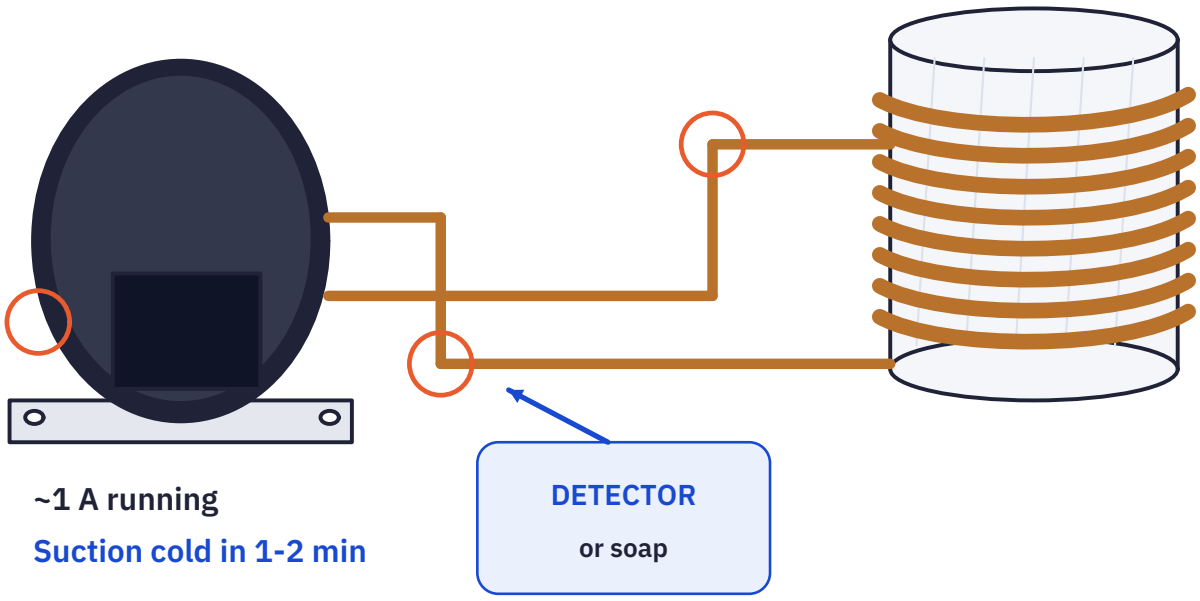
Write down the net grams delivered and the target they were aimed at. The measured charge belongs on the unit and in its record.

DONE WHEN

The BPV31 is closed and capped, clamped on as the service point for life, and the charge is recorded.

Run up and sweep for leaks

First breath: the right current, a cold suction line and silence from the detector at every joint.



Both tie-ins / BPV31 saddle + cap / every thread

3 min off before any restart

1 Energize briefly

Through the inline 5 A fast-blow fuse. Expect ~1 A running; the suction line drops cold within a minute or two.

2 Sweep every joint

Toptes PT520A or soap solution at both tie-ins, the BPV31 saddle and flare cap, and every threaded connection. Zero hits, zero bubbles.

3 Wait 3 minutes to restart

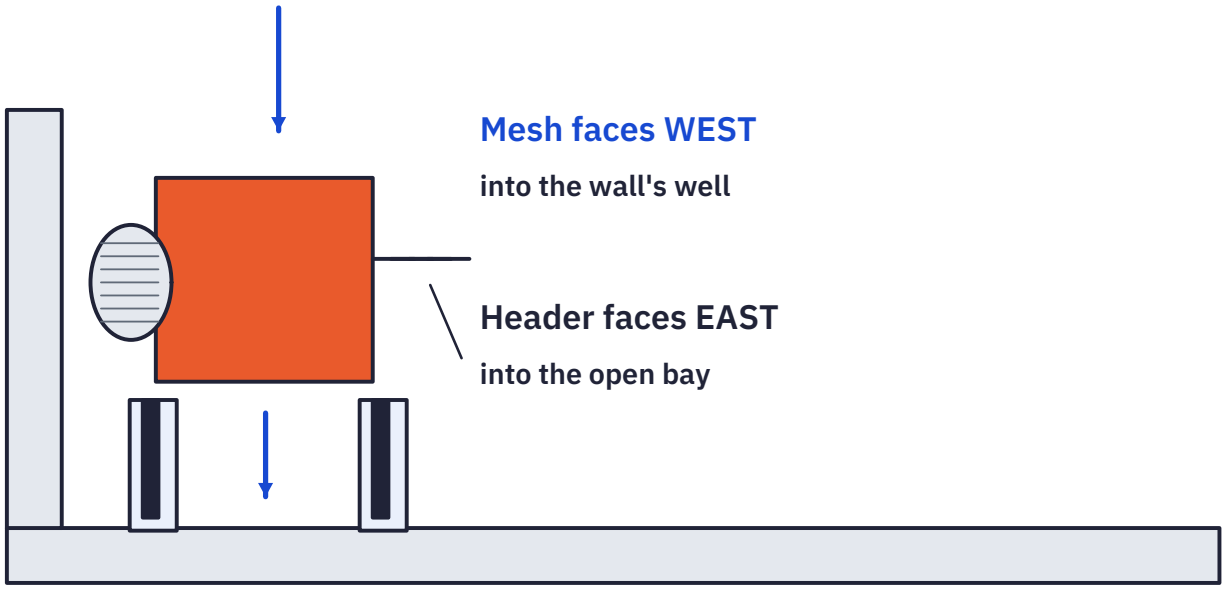
Firmware holds a 3-minute minimum off. A quick restart stalls the motor against head pressure until the overload trips.

DONE WHEN

~1 A draw, suction cold in 1-2 minutes, no detector hit or bubble at any joint.

Place the gas sensor first

Install the MQ-6 in the open floor strip before the compressor makes the slot hard to reach.



Install before the compressor blocks access.

Card is on edge; lower until it lands on its shoulder.

1 Orient the card on edge

The card runs along the west (-X) flank. Mesh faces west into the wall well; its header faces east into the bay.

2 Lower into the grooves

Drop the card straight between the two printed posts until it lands on their shoulder. Route its loom without lifting the card or blocking the low floor zone.

3 Check installed response later

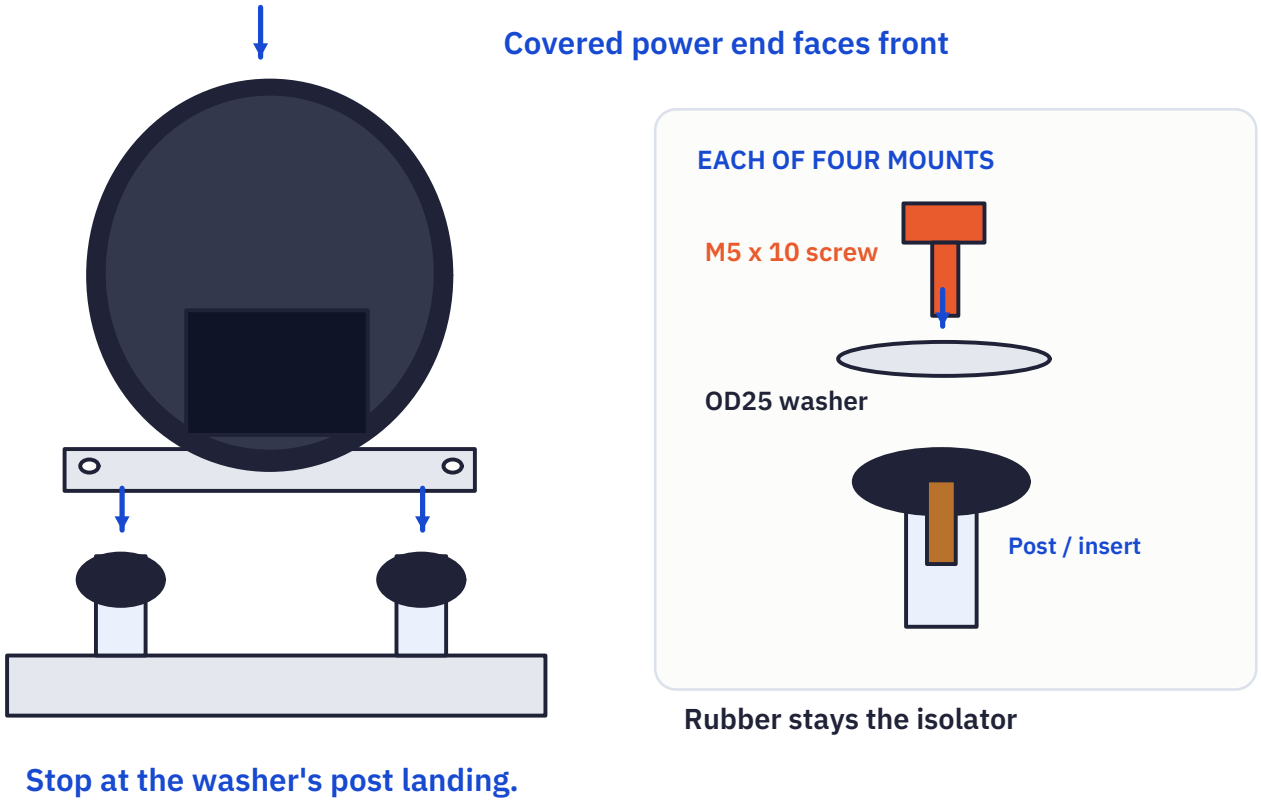
A located sensor is not a commissioned interlock. Prove module polarity, clean-air behavior and actual compressor shutdown under the approved test method.

CHECK

The gas module is reachable, correctly oriented and retained; its threshold/shutdown evidence remains open.

Lower the compressor

Mount the connected refrigeration assembly without transferring its weight into the copper lines.



1 Face the power end forward

Keep the original covered terminal/PTC assembly intact. Support the cold core and condenser while lowering the plate straight onto all four floor posts.

2 Use the four intended stacks

M5 x 10 SHCS with OD25 fender washers, into RX-M5x9.5 inserts. The posts rise through the rubber grommets; keep the grommets seated.

3 Stop at the post crown

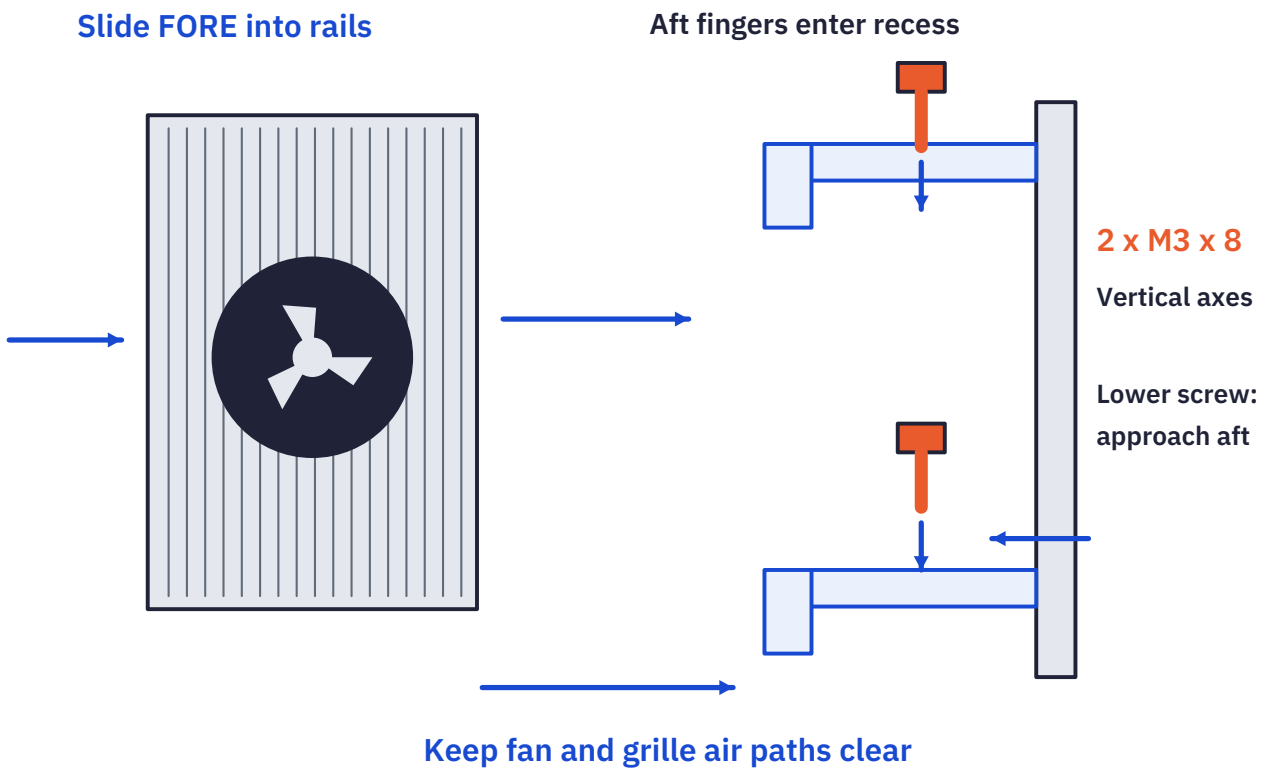
The washer is intended to land on the post after nominal 0.4 mm flange squeeze. Check the real bearing path and isolation; there is no accepted torque or load-life result.

DONE WHEN

All four stacks land without crushing rubber, hanging on tubework or forcing an insert.

Seat the condenser and fan

Use the printed rails and aft fingers to carry the condenser block above the slab.



1 Enter the aft recess

Feed the aft end onto the mounting fingers, then slide fore until the front flanges enter the rail grooves and the face meets their shoulder.

2 Close the two mounts

Drive the two M3 x 8 screws down through the donor's aft flange holes into the fingers' inserts. Approach the lower screw from aft with the bay still open.

3 Keep the air path open

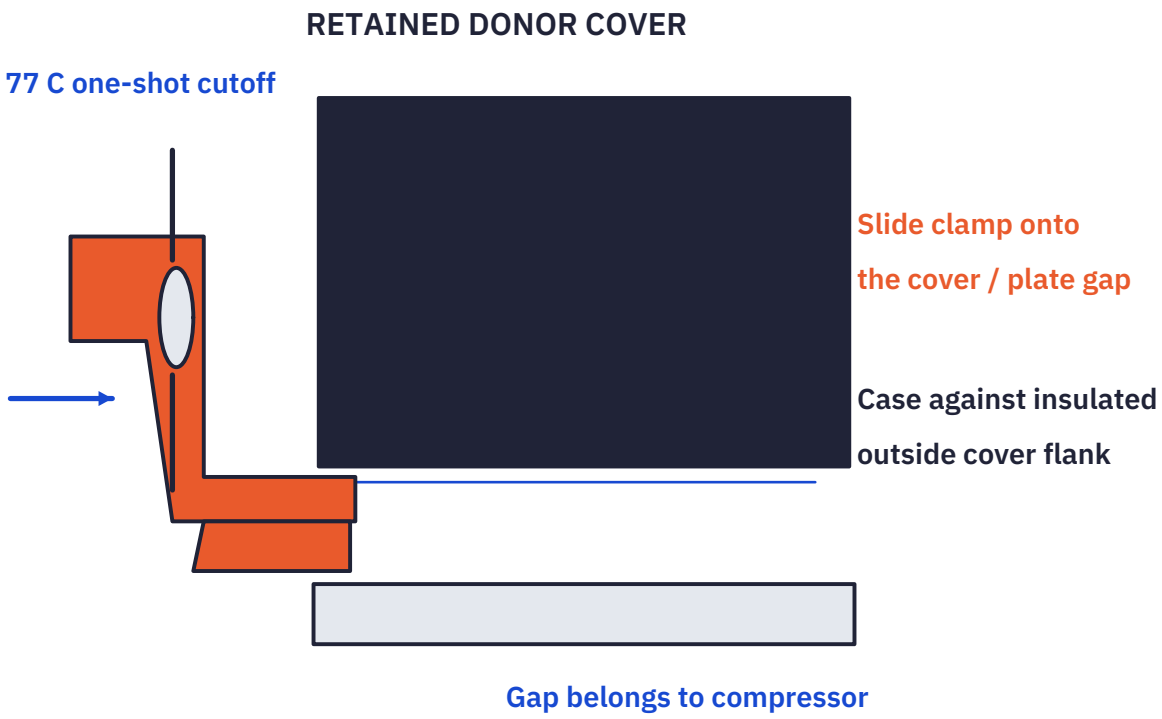
Confirm the native fan's installed flow direction, clear fins and cross-cabinet inlet/exhaust grilles. Inspect every attached line for stress or rubbing.

CHECK

The block is supported by its mounts, the fan turns freely and airflow is confirmed on the real donor.

Seat the thermal cutoff

The 77 C one-shot fuse reads through contact with the outside flank of the retained donor power cover.



Case is live: both contact faces remain insulating.

1 Keep the live case insulated

The cover and printed clamp are the insulating contact surfaces. Keep bare case/leads clear of the compressor can and plate; wiring follows the AC schedule.

2 Place the case and clamp

Seat the fuse along its channel against the cover flank. Slide the clamp's leaves into the cover-to-plate gap so the clamp rides with the compressor.

3 Inspect the actual fit

The fuse must contact the cover without damage or a loose air gap. Nominal CAD diameter does not prove contact across case/print tolerance; preserve required straight lead length.

HOLD

Installed retention, contact and insulation need bench evidence before the cutoff is credited as protection.

Prove the first start

Finish electrical commissioning and leak checks before a brief compressor smoke test.

thermal

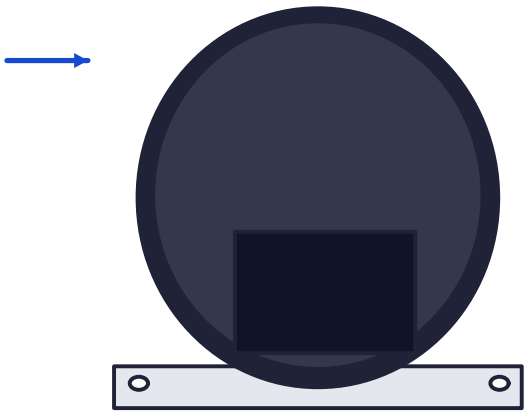
0x28 DS18B20 TANK

0x10 DS18S20 COIL

status

Probe health / gas clear

Relay + fan + off timer



Brief empty-core smoke test

30-60 s under a commissioned control

Observe / stop / honor restart guard

RECORD ACTUAL CURRENT

Compare with donor nameplate

1 Read real probes

Use thermal and status on the shipping firmware. Require family 0x28 on the tank and 0x10 on the coil; de-energized ambient checks are within +/-2 C.

2 Prove the control path

Use a commissioned bench control, with gas interlock and condenser fan working. Honor the 3-minute firmware off guard and any longer donor restart requirement.

3 Observe and stop

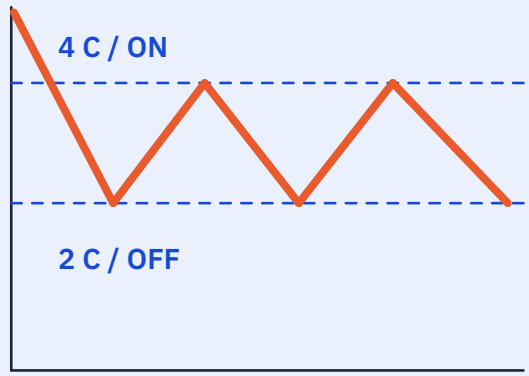
For the empty-core smoke test, 30-60 s is the procedure's brief window. Record actual nameplate-compatible current and falling coil temperature; stop for hum, overload or a leak.

CHECK

This proves switching and probe location only. It does not establish water-load performance or lifetime.

Prove the loaded cold loop

Commission refrigeration with the documented water load and verify the actual control limits.



Carbonator wall

COIL FREEZE GUARD

-8 C / trip

-5 C / recover

3 min minimum off

60 s minimum on

Stale probe / 30 s

Record the actual water-load test

Ambient / fill / charge / temperatures / cycles / current

1 Confirm current firmware values

Tank: on at 4 C, off at 2 C. Coil: trip at -8 C, recover at -5 C. Minimum off 3 min, on 60 s; stale probe after 30 s parks cooling.

2 Run the specified loaded test

Record water fill, ambient, actual charge, starting temperatures, pull-down, outlet-water temperature, coil trend, current and cycle timing.

3 Keep acceptance distinct

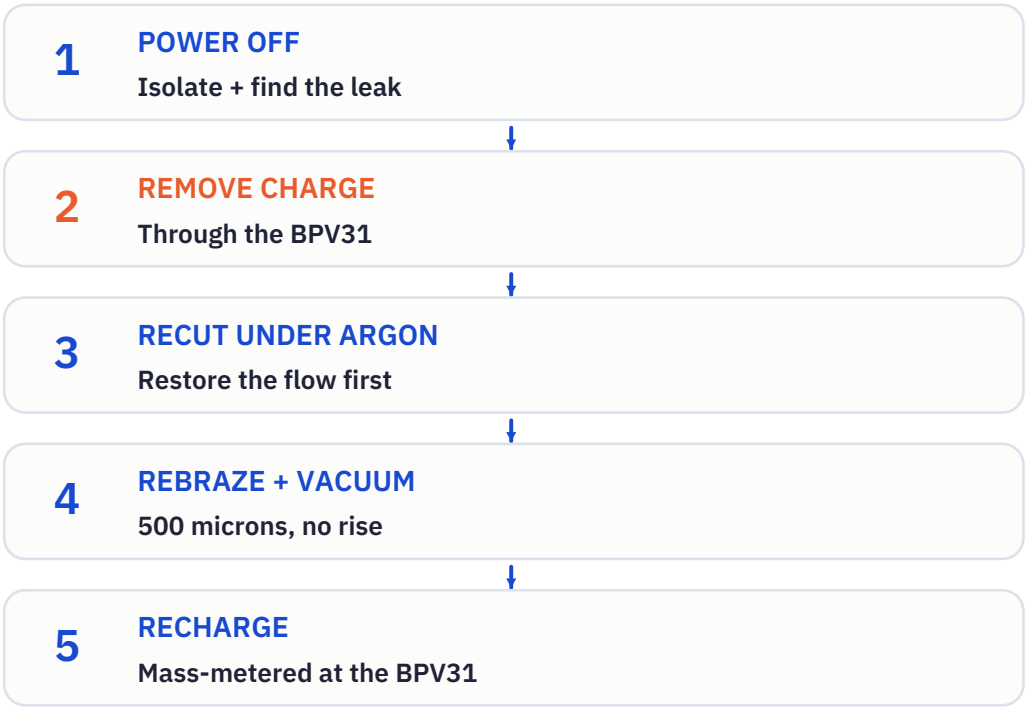
The acceptance document's 8-hour window and 10-70% duty band are proposed defaults. Missing loaded measurements block production acceptance.

DONE WHEN

The logged control behavior and load result meet a committed test specification, with no leak alarm.

Repair through the full sequence

A leak means the full redo: remove the charge, recut, restore argon, rebraze, pull vacuum and recharge.



A charged or leaking circuit never goes directly to heat.

1 Stop and find the leak

Isolate power and ventilate. Find the joint with the detector or soap. A gas alarm, repeated freeze trip or overload is diagnosed, not tuned by guessing.

2 Remove the charge at the BPV31

Use the handling route for the completed dispenser and recovery equipment rated for flammable refrigerants. The donor vent does not cover a rebuilt unit.

3 Redo the sequence

Recut the joint, restore the argon flow from page 11, rebraze with a wet rag, pull vacuum and recharge by mass. Never repair under charge.

CHECK

Any route that applies heat before the charge is out and argon is flowing fails this sequence.

Keep the unit's evidence

The guide supplies order and pictures; the per-unit record establishes which work and measurements were completed.

PER-UNIT REFRIGERATION RECORD

- ☐ **Donor / label / actual topology** — Pages 2-3
- ☐ **Coil / probes / routed protrusions** — Pages 4-8
- ☐ **Vent / argon / brazes / plug faces** — Pages 10-15
- ☐ **Vacuum hold / net charge / leak sweep** — Pages 16-19
- ☐ **Mount / airflow / controls / load test** — Pages 20-25

Name / serial / date / measurements / scope

1 Keep the measured result

Attach donor labels, coil/probe checks, plug-face checks, the vacuum hold, net charge, the leak sweep and installed control/load results.

2 Use the written sources

The footer names the responsible procedure or part reference. The README indexes the local documents.

3 Keep this guide manual

Rebuild it deliberately when its operation changes. The PDFs and source receipt are committed artifacts; the appliance build does not redraw them.

CHECK

Open items stay where refrigerant-loop.md lists them; the unit record shows what was measured.